Supplementary Material

Identifiability of species interactions in a living material: Bayesian calibration of an extended Hamilton principle multiphase model

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S1 Model derivation

State variables and kinematics

Following Klempt et al. (2026, 2024), the biofilm is described at a material point by two sets of internal variables: the volume fraction $\phi_i \in (0,1)$ occupied by species i ($i = 1, \dots, n$; here $n = 5$), and the viability fraction $\psi_i \in (0,1)$ representing the proportion of living cells within species i. The living volume fraction is then

$$\bar{\phi}_i := \phi_i \, \psi_i, \qquad i = 1, \dots, n. \tag{S1}$$

An additional void fraction ϕ_0 closes the system via the holonomic constraint

$$\sum_{l=0}^{n} \phi_l(t) - 1 = 0 \quad \forall \, t. \tag{S2}$$

Variational formulation

The evolution equations result from the stationarity of the rate functional (Junker et al., 2014; Hackl, 1997), the local form of the extended Hamilton principle (Junker and Balzani, 2021; Junker et al., 2025):

$$\dot{\Psi} + \Delta + C \to \mathrm{stat}_{(\dot{\boldsymbol{\xi}}, \gamma)}, \tag{S3}$$

where $\boldsymbol{\xi} = (\boldsymbol{\phi}, \boldsymbol{\psi})^T$ collects the volume fractions and viability fractions, γ is a Lagrange multiplier enforcing the volume constraint, C is the constraint function, and Δ the dissipation function. The free energy density encodes nutrient-driven growth and antibiotic-induced killing:

$$\Psi = -\tfrac{1}{2} \, c^* \, \bar{\boldsymbol{\phi}}^T \mathbf{A} \, \bar{\boldsymbol{\phi}} + \tfrac{1}{2} \, \alpha^* \, \boldsymbol{\psi}^T \mathbf{B} \, \boldsymbol{\psi}, \tag{S4}$$

where c^* is the nutrient concentration, α^* the antibiotic concentration, $\mathbf{A} \in \mathbb{R}^{n \times n}$ the symmetric growth coefficient matrix, and $\mathbf{B} = \mathrm{diag}(b_1, \dots, b_n)$ the antibiotic sensitivity matrix.

The dissipation function is derived from the rate-dependent potential (Klempt et al., 2026):

$$\Delta = \tfrac{1}{2} \, \dot{\bar{\boldsymbol{\phi}}}^T \boldsymbol{\eta} \, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2} \, \dot{\boldsymbol{\phi}}^T \boldsymbol{\eta} \, \dot{\boldsymbol{\phi}}, \tag{S5}$$

with the diagonal viscosity matrix $\boldsymbol{\eta} = \mathrm{diag}(\eta_1, \dots, \eta_n)$. The first term couples $\dot{\phi}_i$ and $\dot{\psi}_i$ through $\dot{\bar{\phi}}_i = \dot{\phi}_i \psi_i + \phi_i \dot{\psi}_i$, giving rise to non-linear dissipative interactions; the second term regularises the system to full rank. Following Klempt et al. (2026), the viscosity parameters are set to $\eta_i = 1$ for all species, rendering them dimensionless scaling factors; the physical time scale is absorbed into the time-step size Δt.

The volume constraint (S2) enters through a Lagrange multiplier γ:

$$C = \gamma \Big(\sum_{l=0}^{n} \phi_l - 1 \Big). \tag{S6}$$

Stationarity conditions

Evaluating the stationarity conditions yields the coupled evolution equations (Klempt et al., 2026):

Volume fraction ($i = 1, \ldots, n$):

$$0 = -c^* \, \psi_i \, (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i(\dot{\phi}_i \, \psi_i^2 + \bar{\phi}_i \, \dot{\psi}_i + \dot{\phi}_i) + \gamma, \tag{S7}$$

Viability fraction ($i = 1, \ldots, n$):

$$0 = -c^* \, \phi_i \, (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \alpha^* \, b_i \, \psi_i + \eta_i(\dot{\psi}_i \, \phi_i^2 + \bar{\phi}_i \, \dot{\phi}_i) + \gamma, \tag{S8}$$

Void fraction:

$$0 = \gamma + \dot{\phi}_0, \tag{S9}$$

Volume constraint:

$$0 = \sum_{l=0}^{n} \phi_l - 1, \tag{S10}$$

In the numerical implementation, a logarithmic barrier penalty Ψ_{bar} with $K_p = \eta_i \times 10^{-4}$ is added to enforce the box constraints $\phi_i, \psi_i \in (0, 1)$ (Klempt et al., 2026); its contribution is omitted from Eqs. (S7)–(S9) for clarity. The system (S7)–(S10) constitutes $2n + 2$ coupled non-linear equations for the unknowns $(\phi_1, \ldots, \phi_n, \phi_0, \psi_1, \ldots, \psi_n, \gamma)$, solved at each time step via Newton's method with an implicit Euler discretisation ($\Delta t = 10^{-4}$, $N_{\text{step}} = 2{,}500$ steps, $t_{\max} = 0.25$; day d is mapped to $t = 0.2375\, d/21$, so day 21 lies at 95% of the integration window).

Only the symmetric part of $\mathbf{A}$ enters the quadratic free energy, so $\mathbf{A}$ is taken symmetric, $a_{ij} = a_{ji}$, and the interactions derive from this potential; the variational derivation ensures thermodynamic consistency (Junker and Balzani, 2021). Since c^* enters the evolution equations only through the product $c^* a_{ij}$, its absolute value is non-identifiable; we fix $c^* = 25$ so that the inferred a_{ij} remain $\mathcal{O}(1)$. The Heine et al. experiments do not involve antibiotic agents, so we set $\alpha^* = 0$, which eliminates the decay term $\alpha^* b_i \psi_i$ from Eq. (S8); the antibiotic sensitivity matrix $\mathbf{B}$ is therefore inactive and excluded from estimation. The symmetric interaction matrix $\mathbf{A}$ has 15 independent entries, with species numbered 1 So, 2 An, 3 Vei, 4 Fn, 5 Pg; these form the 15-dimensional parameter vector $\boldsymbol{\theta}$.

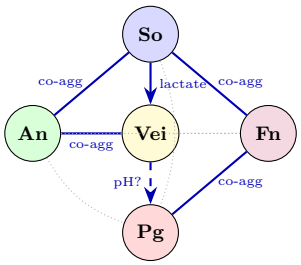
S2 Reported interactions among the five taxa

Figure S1 and Table S1 summarise interactions reported in the literature; they are used only to interpret the inferred coefficients, never as constraints.

S3 Likelihood weights and algorithm

Weight	Value	Applies to
λ_{Pg}	5	Pg, all time points
λ_{late}	3	Days 15 and 21, all species
λ_{rare}	0.1	species with mean fraction $< 5\%$
λ_{ψ}	3.0 (DH), 2.0 (DS), 1.5 (CS, CH)	viability channel, stages (iii)–(iv)

Table S2: Likelihood weights. The composition data are viable-cell (PMA-qRT-PCR) species fractions; the viability channel is the community live fraction from SYTO9/propidium iodide staining.



Pair	Mechanism	Reported flow
So–An	Co-aggregation	–
So–Vei	Lactate exchange	So to Vei
An–Vei	Co-aggregation; Vei uses lactate of So and An	–
So–Fn	Co-aggregation (Fn as bridging organism)	–
Fn–Pg	Co-aggregation (Fn bridges to Pg)	–
Vei–Pg	pH rise from *Veillonella* lactate metabolism (hypothesised in Heine et al., 2025; *Veillonella*-conditioned medium did not change Pg growth)	Vei to Pg (hypothesised)

Table S1: Previously reported or biologically supported interactions (Kolenbrander et al., 2010; Heine et al., 2025) used to interpret the inferred coefficients. The model coefficients a_{ij} are symmetric; the last column refers to the reported metabolite flow (–: physical co-aggregation, no metabolite flow), not to the model. All 15 parameters are estimated; none is fixed a priori.

Figure S1: Interactions considered in interpreting the inferred coefficients, compiled from the literature (Kolenbrander et al., 2010; Heine et al., 2025). Solid: previously reported or biologically supported interaction (labels: reported mechanism; arrows: reported metabolite flow; lines without arrows: co-aggregation; neither is a property of the model, whose matrix $\mathbf{A}$ is symmetric). Dashed: hypothesised, not supported by the conditioned-medium experiment of Heine et al. (2025). Dotted grey: no interaction considered in the interpretation; this does not mean that none exists. The network is not used in the inference: all 15 entries of $\mathbf{A}$ are estimated with the same uniform prior box.

Metric	CPU (12 cores)	GPU (RTX 4090)	Speedup
Matched benchmark (DH, $N_p = 500$, $n_{\text{mut}} = 10$, $K = 9$ tempering steps):			
Per tempering step	2 400 s	14 s	170×
Total	21 600 s	147 s	**147**×

Table S3: Computational cost comparison. The matched benchmark (500 particles, identical settings) isolates the GPU speed-up at about 150× (170× per tempering step, 147× end to end). The cost of every inference stage of the paper is listed in Table S13.

Algorithm S1 Staged GPU-accelerated TMCMC (one seed; every stage is run with three seeds and checked as in the main text)

Require: Composition data $\mathbf{y}_{\text{obs}}$, viability data, N_p particles, GPU device
Ensure: MAP estimate $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, log-evidence $\ln \hat{Z}$

1: ▷ **Stage (i): composition likelihood, initial box**
2: Draw $\{\boldsymbol{\theta}_j\}_{j=1}^{N_p}$ from the stage prior; set $\beta_0 \leftarrow 0$
3: **while** $\beta_m < 1$ **do**
4: Find β_{m+1} s.t. $\mathrm{CoV}[\{w_j\}] = \delta_{\text{target}}$ via bisection
5: Compute weights $w_j \leftarrow p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j)^{\beta_{m+1}-\beta_m}$
6: Resample N_p particles $\propto w_j$ (systematic)
7: Estimate weighted covariance $\hat{\boldsymbol{\Sigma}}^{(m)}$ (weighted particle covariance)
8: **for** $k = 1$ **to** n_{mut} **do**
9: Propose $\boldsymbol{\theta}'_j$ for all j (random-walk or differential-evolution move)
10: Evaluate all N_p likelihoods in parallel via `jax.vmap` on GPU
11: Accept or reject (Metropolis–Hastings); reject proposals outside the box
12: **end for**
13: **end while**
14: $K \leftarrow m$ (number of tempering steps)
15: Record MAP and posterior standard deviation (SD) per parameter
16: ▷ **Stage (ii): composition likelihood, box mean ± 4 SD**
17: Repeat lines 2–14 on the narrowed box, starting from the previous particles
18: ▷ **Stages (iii)–(iv): add the viability channel; initial box, then mean ± 4 SD**
19: Augment likelihood with the weighted viability channel (Table S2)
20: Repeat lines 2–14 for each stage
21: **return** $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, $\ln \hat{Z}$

S4 Prior boxes

Table S4 lists the uniform prior box of every interaction entry for each condition and stage. The stage-(i) boxes were set from earlier calibrations of a previous version of the model on the same data. Some of them exclude negative values or allow only small ones (CS: $a_{11} \in [0, 2]$; DH: $a_{11}, a_{22}, a_{44} \in [0, 5]$, $a_{12} \in [0.5, 3]$, $a_{23} \in [0, 3]$, $a_{45} \in [-0.5, 6]$), so in stages (i)–(ii) the signs of these entries are partly set by the prior; all of these bounds were widened in stage (iii). An entry was widened where more than 20% of its posterior mass lay in the outer 5% of the range on one side, or where this fraction grew by more than 0.10 from the previous stage; the stage was then repeated. The last column gives $\ln V = \sum_k \ln(u_k - l_k)$, the log-volume of the box, which enters $\ln \hat{Z}$ and is the reason why $\ln \hat{Z}$ is not compared across conditions.

S5 Runs and convergence

Table S5 lists, for every condition and stage, the number of particles and mutation steps, the number of tempering steps, the acceptance rate, the moves per parameter, and the spread over the three seeds of the maximum log-likelihood and of the posterior medians (criteria (c)–(e) of the main text). Criteria (a) and (b) are met by all runs. Stages (ii)–(iv) of CS and CH and stages (ii)–(iii) of DS meet all criteria; the DH stage-(iii) group does not yet meet (e) (median spread 0.64 posterior SD, for a_{22}; also a_{35} and a_{45} above 0.5)[prov.]. Stage (i) is exploratory: CS and CH do not meet (c), and DS does not meet (e) because of the two a_{33} modes (Section S8). Criterion (e) is stricter than common practice. Treating the three seeds as chains, the potential scale reduction factor $\hat{R}$ (Gelman and Rubin, 1992) of the identified entries is at most 1.001 for CS and CH (stages (ii)–(iv)) and 1.005 for DS (stage (iii)), and 1.036 for the DH stage-(iii)

Table S4: Prior boxes per condition and stage for the 15 free entries. (i) initial box; (ii) stage-(i) posterior mean ± 4 SD; (iii) back on the box of stage (i), with the bounds widened (bold) where more than 20% of the posterior mass lay in the outer 5% on one side; (iv) stage-(iii) posterior mean ± 4 SD, clipped to the stage-(iii) box (seed with the highest log-likelihood). $\ln V = \sum_k \ln(u_k - l_k)$ over the free entries. **[DS, DH: stage (iv) pending.]**

CS

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[0, 2]	[0, 2]	**[-5, 2]**	[-5, 1.28]
a_{12}	[-0.5, 2]	[-0.18, 2]	[-0.5, 2]	[-0.5, 2]
a_{22}	[-1, 3]	[-1, 3]	[-1, 3]	[-1, 3]
a_{33}	[-1, 1.5]	[-1, 0.95]	**[-15, 20]**	[-15, 6.12]
a_{34}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{44}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{13}	[-1, 3]	[1.17, 3]	**[-1, 20]**	[-1, 13.99]
a_{14}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{23}	[-1, 3]	[-0.71, 3]	[-1, 3]	[-1, 3]
a_{24}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{55}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{15}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{25}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{35}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{45}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
$\ln V$	12.9	11.7	18.5	17.5

CH

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[-1, 1.5]	[-1, 1.5]	**[-5, 1.5]**	[-4.18, 1.5]
a_{12}	[-1, 2.5]	[-0.84, 2.5]	**[-5, 2.5]**	[-5, 2.5]
a_{22}	[-1, 2.5]	[-1, 2.5]	[-1, 2.5]	[-1, 2.5]
a_{33}	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]	[-1, 1.03]
a_{34}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{44}	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]
a_{13}	[-1, 3]	[-0.4, 3]	[-1, 3]	[-1, 2.36]
a_{14}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{23}	[-1, 3]	[-0.85, 3]	[-1, 3]	[-1, 3]
a_{24}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{55}	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]
a_{15}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{25}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{35}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{45}	[-1, 2]	[-1, 2]	[-1, 2]	[-1, 2]
$\ln V$	14.2	14.0	18.1	17.6

DS *(stage (iii) is final, provisional)*

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[-1, 2]	[-1, 2]	[-1, 2]	–
a_{12}	[-1, 1]	[-1, 1]	[-1, 1]	–
a_{22}	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]	–
a_{33}	[-15, 20]	[-15, 15.35]	[-15, 20]	–
a_{34}	[-15, 20]	[-10.47, 11.48]	[-15, 20]	–
a_{44}	[-1, 2.5]	[-1, 2.5]	[-1, 2.5]	–
a_{13}	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]	–
a_{14}	[-1, 1]	[-1, 1]	[-1, 1]	–
a_{23}	[-1, 3]	[-1, 3]	**[-5, 3]**	–
a_{24}	[-0.5, 3]	[-0.5, 3]	[-0.5, 3]	–
a_{55}	[-1, 4]	[-1, 2.49]	**[-5, 4]**	–
a_{15}	[-0.5, 2.5]	[-0.5, 2.5]	**[-5, 2.5]**	–
a_{25}	[-0.5, 2.5]	[-0.5, 2.5]	[-0.5, 2.5]	–
a_{35}	[-3, 3]	[-2.04, 3]	[-3, 3]	–
a_{45}	[-15, 20]	[-2.92, 7.26]	[-15, 20]	–
$\ln V$	24.5	22.1	26.7	–

DH *(stage (iii) is final, provisional)*

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[0, 5]	[0, 5]	**[-5, 5]**	–
a_{12}	[0.5, 3]	[0.5, 3]	**[-5, 3]**	–
a_{22}	[0, 5]	[0, 2.02]	**[-5, 5]**	–
a_{33}	[-1, 3]	[-0.66, 1.73]	**[-15, 20]**	–
a_{34}	[-1, 1]	[-0.28, 1]	**[-15, 20]**	–
a_{44}	[0, 5]	[0, 3.3]	**[-10, 5]**	–
a_{13}	[-3, 3]	[-3, 3]	[-3, 3]	–
a_{14}	[-1.5, 3]	[-1.5, 3]	**[-8, 3]**	–
a_{23}	[0, 3]	[0, 3]	**[-15, 20]**	–
a_{24}	[-0.5, 3]	[-0.5, 3]	**[-15, 20]**	–
a_{55}	[-1.5, 5]	[-1.5, 5]	[-1.5, 5]	–
a_{15}	[-0.5, 2.5]	[-0.5, 2.5]	[-0.5, 2.5]	–
a_{25}	[-0.5, 2.5]	[-0.5, 2.5]	[-0.5, 2.5]	–
a_{35}	[-0.5, 5]	[-0.5, 2.49]	**[-15, 20]**	–
a_{45}	[-0.5, 6]	[-0.5, 6]	**[-5, 20]**	–
$\ln V$	21.1	18.2	38.6	–

group that fails (e) (1.059 over all 15 entries)**[prov.]**. The DH group would therefore pass the conventional threshold $\hat{R} < 1.1$, but not the recommended $\hat{R} < 1.01$ (Vehtari et al., 2021) nor criterion (e); we keep (e) as stated and report the group as not yet accepted. Values from `tools/rhat_seeds.py`. For CS and CH, stage (ii) was started from stage-(i) runs with 120 mutation steps that are not archived; the stage-(i) runs listed here (80 steps) give stage-(ii) bounds within 0.17 of those used.

S6 Identifiability of the interaction entries

Table S7 lists, for the final stage (iv) of every condition (**[DS, DH: stage (iii)]**), the median and the 90% interval of each entry, pooled over the three seeds. The reference box is the prior box of that stage (Table S4, column (iv); for DS and DH column (iii)). An entry whose pooled posterior SD is at least 0.8 times that of a uniform distribution on this box is shown in grey and excluded from the seed-agreement criterion: the data do not constrain its magnitude beyond the prior box. Among these, entries whose 90% interval excludes zero are marked s.o. (sign

Table S5: Runs of the final pipeline (three seeds per row): particles N_p, mutation steps per tempering step n_{mut}, tempering steps K, acceptance rate, moves per free parameter (criterion (c)), maximum log-likelihood (best seed; spread over the three seeds in parentheses, criterion (d)), largest spread of the seed medians over the identified entries in units of the pooled posterior SD (criterion (e), no mode splitting) and wall-clock hours per seed. DS, DH: stage (iii) is the final stage so far **[prov.]**.

Cond.	Stage	N_p	n_{mut}	K	Accept.	Moves/dim	max log L (spread)	Med. spread	Hours
CS	(i)	1000	80	6	0.14–0.14	4.4–4.5	-5.86 (0.11)	0.11	2.9–3.0
CS	(ii)	2000	120	6	0.14–0.14	6.7–6.8	-5.67 (0.27)	0.14	4.7–5.1
CS	(iii)	2000	160	6–7	0.13–0.14	8.8–9.9	-16.06 (0.19)	0.10	5.3–7.5
CS	(iv)	5000	150	6–7	0.13–0.13	7.9–9.2	-15.81 (0.24)	0.13	7.7–9.0
CH	(i)	1000	80	5	0.14–0.14	3.8–3.9	-3.19 (0.10)	0.07	2.5–2.5
CH	(ii)	2000	150	4	0.15–0.15	5.9–6.0	-3.15 (0.08)	0.05	3.9–4.2
CH	(iii)	2000	150	7	0.13–0.13	8.9–9.1	-8.86 (0.09)	0.07	7.4–7.5
CH	(iv)	5000	150	7	0.13–0.13	8.9–9.0	-8.65 (0.11)	0.07	9.7–9.8
DS	(i)	1000	80	10	0.11–0.12	5.9–6.2	-0.91 (0.72)	1.55	5.2–5.4
DS	(ii)	3000	100	10	0.12–0.12	8.0–8.2	-0.58 (0.27)	0.09	5.7–6.4
DS	(iii)	18000	150	10	0.11–0.12	11.4–11.6	-6.45 (0.53)	0.45	15.8–17.6
DH	(i)	1000	80	10	0.12–0.13	6.6–6.7	-3.92 (0.51)	0.19	4.9–5.1
DH	(ii)	2000	80	8–9	0.12–0.13	5.4–6.0	-3.21 (0.31)	0.21	4.3–5.2
DH	(iii)	14000	120	12	0.10–0.10	9.9–10.1	-14.65 (0.16)	0.64	15.3–16.6

only), the others n.i. (not identified). All other entries are called identified (constrained by the data); an identified entry need not be sign-determined (e.g. a_{45} in DS and DH). Where the stage-(iv) box was narrowed to the stage-(iii) mean ± 4 SD, the uniform SD on that box is about 2.3 stage-(iii) SDs; for these entries the classification is in effect made against the stage-(iii) box.

The threshold 0.8 is a convention. Table S6 counts the identified entries for thresholds between 0.5 and 0.9. The counts change, most in the commensal conditions, where some entries lie close to 0.8 (CS: 0.77 and 0.81; CH: 0.73 and 0.83), whereas DS has no entry between 0.71 and 0.93. The pattern does not change: at every threshold the dysbiotic conditions have at least as many identified entries as the commensal ones, and the entries identified in the commensal conditions are interactions of So, An and Vei; at 0.9 they also include So–Fn and So–Pg, the two entries that are sign-determined in CH.

Table S6: Number of identified entries of **A** (of 15) for different thresholds on the ratio of the pooled posterior SD to the SD of a uniform distribution on the prior box; *value used throughout. **[DS, DH: stage (iii).]** Generated by `tools/ident_threshold_table.py`.

Threshold	0.5	0.7	0.8*	0.9
CS	1	3	4	7
CH	2	5	6	8
DS	3	7	8	8
DH	7	8	9	12

Curvature of the likelihood. As a check that does not use the posterior samples, we computed the Hessian **H** of $-\ln p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta})$ at the MAP over the 15 free entries, scaled each entry by the standard deviation of a uniform distribution on its box, and counted the eigen-directions with scaled curvature of at least $1/0.8^2$, i.e. directions in which the likelihood is narrower than the identification threshold. Their number is 4, 4, 8 and 5 in CS, CH, DS and DH, against 4, 6, 8 and 9 entries identified from the posterior **[prov.]**. The counts agree in CS and DS; in CH and DH the posterior identifies more entries than the local curvature, because several entries are constrained jointly along correlated directions and by the non-quadratic shape of the likeli-

Table S7: Posterior median [5%, 95%] of the 15 entries at stage (iv) ([DS, DH: stage (iii)]), three seeds pooled. Grey: the posterior spreads over most of the prior box of the stage (SD ≥ 0.8 times that of a uniform distribution on the box); n.i.: not identified, s.o.: sign only (the 90% interval excludes zero).

Entry	CS	CH	DS	DH
a_{11}	-3.58 [-4.9, -1.2]	-0.99 [-2.5, -0.5]	$+0.02$ [-0.9, $+1.7$] n.i.	-0.85 [-4.6, $+4.3$] n.i.
a_{12}	$+0.77$ [-0.3, $+1.8$] n.i.	-0.44 [-3.5, $+0.6$]	-0.07 [-0.9, $+0.9$] n.i.	-1.35 [-4.6, $+2.3$] n.i.
a_{22}	$+0.68$ [-0.7, $+2.2$]	-0.13 [-0.9, $+1.5$]	$+0.14$ [-0.9, $+1.3$] n.i.	-2.98 [-4.7, -0.7]
a_{33}	-6.76 [-13.4, -2.5]	-0.53 [-1.0, $+0.2$]	-0.63 [-1.5, -0.3]	-1.28 [-2.8, -0.7]
a_{34}	-0.29 [-0.9, $+0.8$] n.i.	-0.20 [-0.9, $+0.8$] n.i.	-2.25 [-12.3, $+0.5$]	-1.29 [-4.4, $+3.7$]
a_{44}	-0.01 [-0.9, $+0.9$] n.i.	$+0.19$ [-0.9, $+1.4$] n.i.	-0.12 [-0.9, $+2.0$] n.i.	-6.88 [-9.7, -2.4]
a_{13}	$+5.74$ [$+2.8$, $+9.2$]	$+0.61$ [$+0.0$, $+1.4$]	-0.16 [-0.9, $+0.7$]	-1.00 [-2.7, $+1.1$]
a_{14}	-0.40 [-0.9, $+0.6$] n.i.	-2.89 [-4.8, -0.2] s.o.	-0.10 [-0.9, $+0.9$] n.i.	-3.19 [-7.0, $+1.3$] n.i.
a_{23}	$+0.50$ [-0.8, $+2.5$] n.i.	$+0.98$ [-0.1, $+2.0$]	-1.84 [-4.5, -0.1]	$+1.18$ [-2.1, $+2.9$]
a_{24}	-0.22 [-0.9, $+0.8$] n.i.	-0.13 [-0.9, $+0.8$] n.i.	$+1.16$ [-0.3, $+2.8$] n.i.	$+5.10$ [$+2.7$, $+7.1$]
a_{55}	-0.04 [-0.9, $+0.9$] n.i.	$+0.18$ [-0.9, $+1.4$] n.i.	-2.09 [-4.7, -0.3]	$+0.97$ [-1.2, $+3.8$] n.i.
a_{15}	-0.42 [-0.9, $+0.6$] n.i.	-3.12 [-4.8, -0.3] s.o.	-1.45 [-4.3, $+0.5$]	$+0.99$ [-0.2, $+2.3$] n.i.
a_{25}	-0.23 [-0.9, $+0.8$] n.i.	-0.20 [-0.9, $+0.8$] n.i.	$+1.18$ [-0.3, $+2.4$] n.i.	$+1.10$ [-0.2, $+2.3$] n.i.
a_{35}	-0.32 [-0.9, $+0.7$] n.i.	-0.29 [-0.9, $+0.8$] n.i.	$+1.04$ [$+0.3$, $+2.4$]	-6.55 [-13.4, -1.7]
a_{45}	-0.05 [-0.9, $+0.9$] n.i.	$+0.45$ [-0.9, $+1.8$] n.i.	$+2.81$ [-1.3, $+15.1$]	$+1.21$ [-1.9, $+7.5$]

hood, which a single Hessian at the MAP does not capture. The MAP is the best particle and not an exact optimum (4–5 small negative eigenvalues), so entry-wise Laplace intervals are not reported.

S7 MAP and posterior bulk by seed

Figure S3 overlays the MAP of each seed on the posterior predictive band pooled over the three seeds. In all conditions the three MAPs nearly coincide. In CS the MAP lies apart from the bulk of the posterior for So and Vei: the pointwise posterior median has a composition error of 0.125, the MAP of 0.179; in the other conditions the two differ by less than 0.02[prov.].

S8 Bayes factor for the *Fn*–*Pg* interaction

Stage (i) was repeated with $a_{45} = 0$ for all four conditions (three seeds each, except CH, for which only one seed is archived). Stage (i) is exploratory and not all of its groups meet the acceptance criteria (Section S5), so these Bayes factors are indicative. Table S8 reports $\ln \hat{Z}$ of both models, $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$, and the width of the a_{45} box, which sets the Occam penalty.

Table S8: Log-evidence of the full and the reduced ($a_{45} = 0$) model at stage (i), mean ± half-range over three seeds (CH, $a_{45} = 0$: one seed), and $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$. The last two columns give the a_{45} box and its log-width, which enters $\ln \hat{Z}_{\text{full}}$ as an Occam penalty.

Cond.	$\ln \hat{Z}_{\text{full}}$	$\ln \hat{Z}_{a_{45}=0}$	$\ln B$	a_{45} box	ln(width)
CS	-15.16 ± 0.08	-15.26 ± 0.13	$+0.10$	[-1, 1]	0.69
CH	-9.62 ± 0.12	-9.66 (one seed)	$+0.05$	[-1, 2]	1.10
DS	-21.48 ± 0.70	-18.81 ± 0.51	-2.67	[-15, 20]	3.56
DH	-21.36 ± 0.38	-23.30 ± 0.34	$+1.94$	[-0.5, 6]	1.87

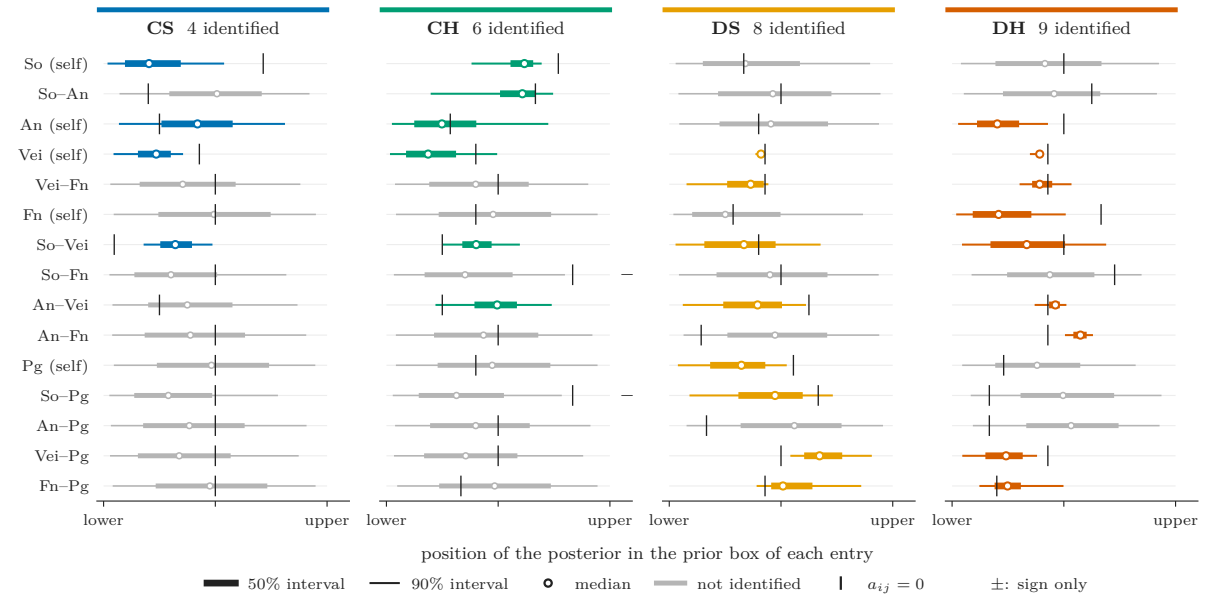


Figure S2: Marginal posteriors of the 15 entries of **A** at the final stage (three seeds pooled), drawn as position within the prior box of the final stage of each entry (Table S4): thick bar 50%, thin bar 90% interval, circle median; black tick: $a_{ij} = 0$. Coloured: identified; grey: not identified; a sign at the right: only the sign is determined (classification as in Fig. 3 of the main text). **[DS, DH from stage (iii).]**

Dysbiotic static. In DS, a_{45} and a_{33} trade off at stage (i). With $a_{45} = 0$ the posterior moves almost entirely to the mode $a_{33} \approx -10$ (weight 0.83–0.87 across seeds), whereas the full model splits its mass between $a_{33} \approx +1$ and $a_{33} < -5$, with a weight of the latter that differs strongly between seeds (0.14–0.66); this stage-(i) group does not pass criterion (e) and is used here only for the Bayes factor. In the mode $a_{33} \geq -5$ the coefficient a_{45} is larger than in the mode $a_{33} < -5$ (Table S9). At the final stage the mode $a_{33} < -5$ is empty in all three seeds, and a_{45} has median +2.8 with a 90% interval reaching zero ($[-1.3, +15.1]$)**[prov.]**.

Table S9: DS, stage (i), full model: a_{45} by a_{33} mode (three seeds pooled, 3000 particles).

Mode	Particles	a_{45} median [5%, 95%]	max log-likelihood
$a_{33} \geq -5$	1761	+2.4 [+0.6, +4.7]	−0.91
$a_{33} < -5$	1239	+0.8 [−0.3, +1.6]	−1.49

S9 Effect of the likelihood weights

Stage (i) for DH was repeated with $\lambda_{\mathrm{Pg}} = \lambda_{\mathrm{late}} = 1$ and $\lambda_{\mathrm{rare}} = 0.1$ unchanged (1000 particles, 100 mutation steps, three seeds; all convergence criteria met). Table S10 compares it with the weighted run.

S10 Viable and total-count composition

The species fractions used in the inference are viable-cell fractions from quantitative real-time PCR with propidium monoazide (PMA) pre-treatment (Fig. 3 of Heine et al., 2025; medians read off the published box plots, nine replicates per time point; the replicate values themselves were

Table S10: DH, stage (i): weighted ($\lambda_{\mathrm{Pg}} = 5$, $\lambda_{\mathrm{late}} = 3$) and unweighted likelihood. Ranges over three seeds; RMSE: root-mean-square error of the composition; the measured Pg ratio of day 21 to day 15 is 2.74.

	a_{45} median [5%, 95%]	Pg D21/D15	RMSE
Weighted	+2.9 to +3.2 [+0.5 to +1.1, +5.2 to +5.5]	2.4–2.9	0.058–0.065
Unweighted	+3.0 [−0.06 to +0.17, +5.6]	1.7–3.1	0.059–0.068

not available). The corresponding total-count fractions without PMA (their Fig. S1) differ little from them: the RMSE between the two sets of medians is 0.057 (CS), 0.042 (CH), 0.033 (DS) and 0.072 (DH). We therefore assume in the composition channel that viability does not depend on the species and compare the data with the normalised volume fractions $\phi_i / \sum_j \phi_j$. The viability channel is the community live fraction from SYTO9/propidium iodide (LIVE/DEAD BacLight) staining, which is image based and not resolved by species. Table S11 compares the final MAP with both data sets; the model is about as close to the total-count data as to the viable data used in the fit.

Table S11: RMSE of the normalised MAP volume fractions $\phi_i / \sum_j \phi_j$ of the final stage against the medians of the viable-cell data used in the fit (PMA-qRT-PCR) and of the total-count data (Heine et al., 2025, Fig. S1).

	CS	CH	DS	DH
Viable (used in the fit)	0.179	0.076	0.027**[prov.]**	0.083**[prov.]**
Total count	0.136	0.059	0.036**[prov.]**	0.045**[prov.]**

S11 Species-resolved viability implied by the data

The viable and the total-count fractions together imply a viability for each species, $\psi_i^{\mathrm{data}} = (y_i^{\mathrm{viable}} / y_i^{\mathrm{total}})\, \ell$, where y_i are the normalised fractions and ℓ is the community live fraction (medians read off the box plots; days with a total-count fraction below 2% are left out). Table S12 compares it with the viability ψ_i of the final MAP. In the data, the viabilities of most species lie close to the community live fraction; the exception are Fn and Pg in the commensal community, which are mostly dead ($\psi^{\mathrm{data}} \approx 0.04$–$0.10$) but make up only a few percent of the total count. The assumption of a species-independent viability in the composition channel is therefore violated mainly for these two rare species, which is why the viable and the total-count compositions differ little (Section S10). The model viabilities do not reproduce ψ^{data}: they are close to it in CS but lie near 0 or 1 for several species in the other conditions, because the likelihood constrains only the community live fraction. Using the two compositions as separate channels—the viable fractions against $\bar{\phi}_i / \sum_j \bar{\phi}_j$ and the total-count fractions against $\phi_i / \sum_j \phi_j$—would constrain ψ_i species by species; this is not done here and is left to future work.

S12 Composition-only fit in CS

To separate a limitation of the model from the conflict between the two likelihood channels in CS, we minimised the composition error alone (unweighted mean squared difference of the normalised fractions; viability channel, likelihood weights and prior omitted) with a gradient-based local optimiser (L-BFGS-B), started from the final MAP and from two random perturbations of it, with every entry restricted to $[-15, 20]$. The best solution reaches a composition RMSE of 0.075 (per species 0.13, 0.04, 0.10, 0.005 and 0.005 for So, An, Vei, Fn and Pg), against 0.179 at the

Table S12: Viability of each species: implied by the data, ψ_i^{data} (median over days 3–21), and of the final MAP, ψ_i (median over the same days). Generated by `tools/psi_from_data.py`. **[DS, DH: stage (iii).]**

	So		An		Vei		Fn		Pg	
	data	model	data	model	data	model	data	model	data	model
CS	0.81	0.91	0.69	0.90	0.64	0.62	0.06	0.08	0.10	0.09
CH	0.42	0.26	0.37	0.90	0.39	0.92	0.04	0.06	0.05	0.06
DS	0.37	0.22	0.57	0.07	0.57	0.41	0.42	0.88	0.52	0.90
DH	0.44	0.09	0.71	0.93	0.34	0.26	0.48	0.91	0.29	0.07

MAP, but its community live fraction is 0.97 at every time point, against the measured 0.65–0.77. Maximising the full likelihood in the same way from the MAP lowers the composition RMSE only to 0.166. The model can therefore follow the rapid rise of So between day 1 and day 3, but not together with the measured live fraction. This is a deterministic diagnostic, not a posterior; its solution lies partly outside the prior box (e.g. $a_{13} = 20$).

S13 Growth-induced stress at a constrained material point

To show how the calibration enters a mechanical model, the posterior dynamics of each condition drive the isotropic growth law of the continuum model of Klempt et al. (2024): $\mathbf{F}_g = (1+\alpha)\mathbf{I}$ with $\mathrm{d}\alpha/\mathrm{d}T^* = k_\alpha \varphi_g$, where T^* is time normalised by the length of the experiment (21 days) and $k_\alpha = 10^{-3}$ as in that work. At a material point that is constrained in all directions, $\mathbf{F} = \mathbf{I}$, the elastic part is $\mathbf{F}_e = \mathbf{I}/(1+\alpha)$ with $J_e = (1+\alpha)^{-3}$, the deviatoric stress vanishes and the Cauchy stress is $\boldsymbol{\sigma} = K(J_e - 1)\mathbf{I}$, with $K = E/[3(1-2\nu)]$, $E = E_0(\varphi_{\mathrm{tot}}^2 + f)$, $\nu = 0.49$ and $f = 10^{-3}$, so no finite-element solution is needed. We report the growth pressure $p = -\frac{1}{3}\operatorname{tr}\boldsymbol{\sigma}$ relative to E_0 for two choices of the driver φ_g: the living volume $\sum_i \varphi_i \psi_i$, whose community fraction is constrained by the live/dead data, and the total volume $\varphi_{\mathrm{tot}} = \sum_i \varphi_i$, which the likelihood does not constrain (it uses fractions only). Driven by the total volume, the four conditions are indistinguishable: φ_{tot} stays at 0.95–0.97 and p/E_0 on day 21 is 0.042–0.046 in all of them (Fig. S4b, dashed). Driven by the living volume, they separate: p/E_0 on day 21 is 0.020 (90% interval $[0.017, 0.023]$) in CH, 0.024 ($[0.021, 0.027]$) in DS and DH, and 0.041 ($[0.032, 0.044]$; MAP 0.030) in CS**[prov.]**. The high CS value follows the bulk of the CS posterior, whose live fraction (0.92 on day 21) lies above the measured 0.65–0.77 (Section S12); the CS MAP, which matches the live fraction better, gives 0.030. This is an illustration of the coupling, not a prediction of the stress in a biofilm: k_α and E_0 are taken from another system and p scales linearly with both, and a spatial model adds nutrient transport, boundary conditions and the spatial distribution of the species. It shows that the dependence of the growth stress on the condition enters through the calibrated viability rather than through assumed species-specific growth parameters, and that the posterior spread can be propagated to the stress.

References

Gelman, A., Rubin, D.B., 1992. Inference from iterative simulation using multiple sequences. Statistical Science 7, 457–472. doi:10.1214/ss/1177011136.

Hackl, K., 1997. Generalized standard media and variational principles in classical and finite elastoplasticity. Journal of the Mechanics and Physics of Solids 45, 667–688. doi:10.1016/S0022-5096(96)00110-X.

Heine, N., Bittroff, K., Szafrański, S.P., Duitscher, M., Behrens, W., Vollmer, C., Mikolai, C.,

Kommerein, N., Debener, N., Frings, K., Heisterkamp, A., Scheper, T., Torres-Mapa, M.L., Bahnemann, J., Stiesch, M., Doll-Nikutta, K., 2025. Influence of species composition and cultivation condition on peri-implant biofilm dysbiosis in vitro. Frontiers in Oral Health 6, 1649419. doi:10.3389/froh.2025.1649419.

Junker, P., Balzani, D., 2021. An extended Hamilton principle as unifying theory for coupled problems and dissipative microstructure evolution. Continuum Mechanics and Thermodynamics 33, 1931–1956. doi:10.1007/s00161-021-01017-z.

Junker, P., Bode, T., Hackl, K., 2025. On a holistic variational formulation for material modeling including dissipative evolution. Journal of the Mechanics and Physics of Solids 200, 106133. doi:10.1016/j.jmps.2025.106133.

Junker, P., Makowski, J., Hackl, K., 2014. The principle of the minimum of the dissipation potential for non-isothermal processes. Continuum Mechanics and Thermodynamics 26, 259–268. doi:10.1007/s00161-013-0299-4.

Klempt, F., Geisler, H., Soleimani, M., Junker, P., 2026. A continuum multi-species bacterial growth model with a novel interaction scheme. Archive of Applied Mechanics 96, 164. doi:10.1007/s00419-026-03160-y.

Klempt, F., Soleimani, M., Wriggers, P., Junker, P., 2024. A Hamilton principle-based model for diffusion-driven biofilm growth. Biomechanics and Modeling in Mechanobiology 23, 2091–2113. doi:10.1007/s10237-024-01883-x.

Kolenbrander, P.E., Palmer Jr., R.J., Periasamy, S., Jakubovics, N.S., 2010. Oral multispecies biofilm development and the key role of cell–cell distance. Nature Reviews Microbiology 8, 471–480.

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., Bürkner, P.C., 2021. Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC. Bayesian Analysis 16, 667–718. doi:10.1214/20-BA1221.

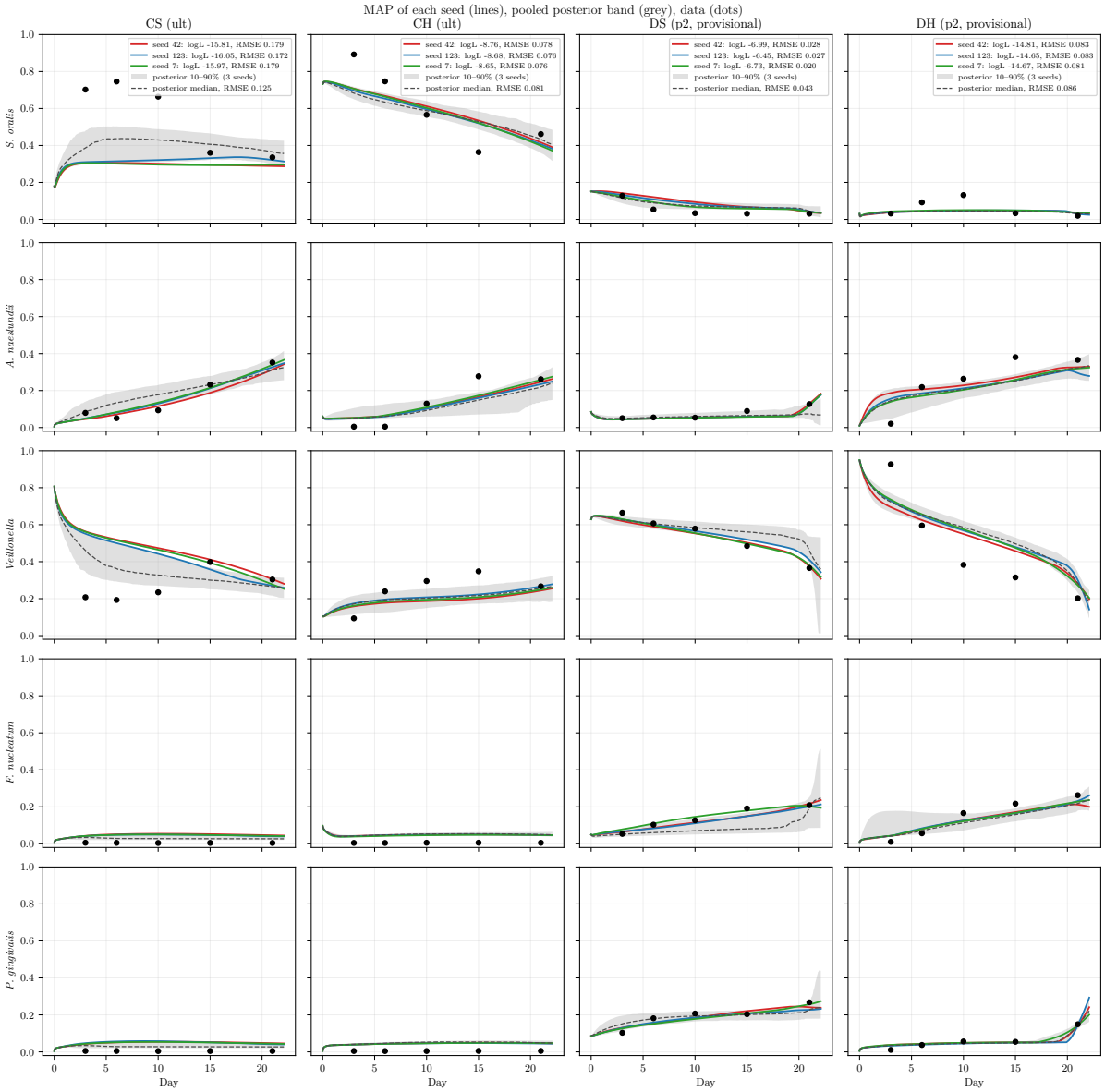


Figure S3: Species fractions: MAP of each seed (coloured lines), pooled central 80% posterior predictive band (10th–90th percentile, grey) and its pointwise median (dashed), and data (dots). **[DS, DH from stage (iii).]**

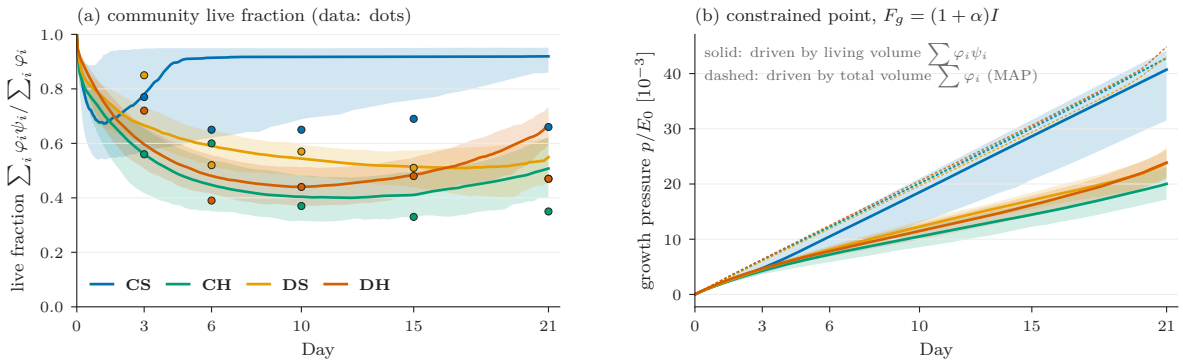


Figure S4: Growth-induced stress at a constrained material point. (a) Community live fraction $\sum_i \varphi_i\psi_i / \sum_i \varphi_i$ (line: posterior median, band: central 90%, 200 posterior samples per condition; dots: live/dead data used in the inference). (b) Growth pressure p/E_0 for growth driven by the living volume (solid, with band) and by the total volume (dashed, MAP). Generated by `tools/growth_stress_check.py`. **[DS, DH from stage (iii).]**

Cond.	Stage	N_p	n_{mut}	K	Wall time / seed	N_{lik}
CS	(i)	1000	80	6	2.9 h	4.8×10^5
	(ii)	2000	120	6	4.7 h	1.4×10^6
	(iii)	2000	160	6	6.7 h	1.9×10^6
	(iv)	5000	150	7	8.9 h	5.3×10^6
CH	(i)	1000	80	5	2.5 h	4.0×10^5
	(ii)	2000	150	4	4.0 h	1.2×10^6
	(iii)	2000	150	7	7.4 h	2.1×10^6
	(iv)	5000	150	7	9.7 h	5.3×10^6
DS	(i)	1000	80	10	5.3 h	8.0×10^5
	(ii)	3000	100	10	5.7 h	3.0×10^6
	(iii)*	18000	150	10	17.6 h	2.7×10^7
DH	(i)	1000	80	10	5.0 h	8.0×10^5
	(ii)	2000	80	9	4.6 h	1.4×10^6
	(iii)*	14000	120	12	16.6 h	2.0×10^7

Table S13: Cost of each inference stage (median over the three seeds): particle number N_p, mutation steps per tempering step n_{mut}, number of tempering steps K, wall time per seed on one GPU, and number of likelihood evaluations $N_{\mathrm{lik}} = N_p(1 + \sum n_{\mathrm{mut}})$. *DS, DH: stage (iii), until stage (iv) is accepted **[prov.]**. Generated by `tools/make_cost_table.py`.